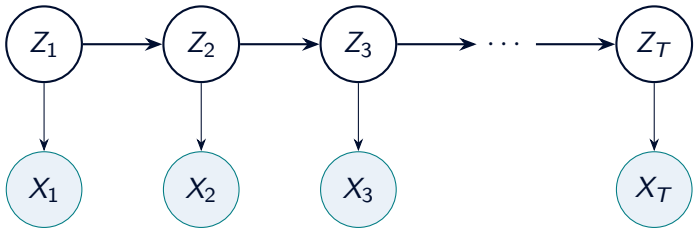
Hidden Markov models

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The hidden-state model



- Hidden states: $Z_t \in \{1, \ldots, K\}$.
- Observed responses: $x_{1:T} = (x_1, \ldots, x_T)$.

The Markov property

Given the previous state, earlier states add no information:

$$\mathbb{P}_\theta(Z_t = j \mid Z_{1:t-1}) = \mathbb{P}_\theta(Z_t = j \mid Z_{t-1}).$$

The assumption is conditional independence.

Initial and transition laws

Initial state:

$$\pi_i = \mathbb{P}(Z_1 = i), \qquad \sum_i \pi_i = 1.$$

Time-homogeneous transitions:

$$a_{ij} = \mathbb{P}(Z_{t+1} = j \mid Z_t = i), \qquad \sum_j a_{ij} = 1.$$

Transition matrix (2×2):

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}.$$

The emission model

The current state determines the response distribution:

$$b_k(x_t; \eta_k) = p_\theta(x_t \mid Z_t = k).$$

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Given the entire state sequence, responses are independent:

$$p_\theta(x_{1:T} \mid z_{1:T}) = \prod_{t=1}^{T} b_{z_t}(x_t; \eta_{z_t}).$$

A density for Gaussian data; a mass function for counts.

The joint likelihood

$$p_\theta(z_{1:T}) = \pi_{z_1} \prod_{t=1}^{T-1} a_{z_t z_{t+1}}, \qquad p_\theta(x_{1:T} \mid z_{1:T}) = \prod_{t=1}^{T} b_{z_t}(x_t; \eta_{z_t}).$$

$$\boxed{p_\theta(x_{1:T}, z_{1:T}) = \pi_{z_1} \prod_{t=1}^{T-1} a_{z_t z_{t+1}} \prod_{t=1}^{T} b_{z_t}(x_t).}$$

The complete-data log-likelihood separates:

$$\ell_c(\theta; x, z) = \log \pi_{z_1} + \sum_{t=1}^{T-1} \log a_{z_t z_{t+1}} + \sum_{t=1}^{T} \log b_{z_t}(x_t; \eta_{z_t}).$$

The marginal likelihood

The state sequence is unobserved, so sum over all paths:

$$p_\theta(x_{1:T}) = \sum_{z_1=1}^{K} \cdots \sum_{z_T=1}^{K} p_\theta(x_{1:T}, z_{1:T}).$$

The maximum likelihood target is

$$\ell(\theta) = \log p_\theta(x_{1:T}), \qquad \widehat{\theta} \in \arg\max_\theta \ell(\theta).$$

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1. **Evaluation:** a direct sum contains K^T paths.
2. **Optimization:** the log of a sum couples the parameters.

Recall the EM expectation

Use t for time and m for the EM iteration.

$$\mathbb{E}_m[g(x, Z)] = \sum_{z_{1:T}} p_{\theta^{(m)}}(z_{1:T} \mid x_{1:T})\, g(x, z).$$

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$$\mathbb{E}_m[g(x, Z)] = \sum_{z_{1:T}} p_{\theta^{(m)}}(z_{1:T} \mid x_{1:T})\, g(x, z).$$

$$Q(\theta \mid \theta^{(m)}) = \mathbb{E}_m[\log p_\theta(x_{1:T}, Z_{1:T})].$$

Hold the $\theta^{(m)}$ fixed when optimizing θ.

Rewrite joint log-likelihood

Introduce the indicators

$$I_t(k) = \mathbf{1}\{Z_t = k\}, \quad J_t(i,j) = \mathbf{1}\{Z_t = i, Z_{t+1} = j\}$$

Rewrite the same complete-data log-likelihood as

$$\begin{aligned}\ell_c(\theta; x, Z) = &\sum_{k=1}^{K} I_1(k) \log \pi_k \\ &+ \sum_{t=1}^{T-1} \sum_{i=1}^{K} \sum_{j=1}^{K} J_t(i,j) \log a_{ij} \\ &+ \sum_{t=1}^{T} \sum_{k=1}^{K} I_t(k) \log b_k(x_t; \eta_k)\end{aligned}$$

State responsibilities

Define the indicator

$$I_t(k) = \mathbf{1}\{Z_t = k\}.$$

Define the state responsibilities:

$$\gamma_t^{(m)}(k) := \mathbb{P}_{\theta^{(m)}}(Z_t = k \mid x_{1:T}) = \mathbb{E}_m[I_t(k)].$$

$\gamma_t^{(m)}(k)$ **is the posterior weight of state** k **at time** t**.**

Transition responsibilities

Define the indicator

$$J_t(i,j) = \mathbf{1}\{Z_t = i, Z_{t+1} = j\}$$

Define

$$\xi_t^{(m)}(i,j) := \mathbb{P}_{\theta^{(m)}}(Z_t = i, Z_{t+1} = j \mid x_{1:T}) = \mathbb{E}_m[J_t(i,j)].$$

This is the expected indicator of an $i \to j$ transition.

$$\xi_t^{(m)}(i,j) \neq \gamma_t^{(m)}(i)\gamma_{t+1}^{(m)}(j) \quad \text{in general.}$$

Adjacent states remain dependent in the posterior.

Q uses expected counts

$$Q(\theta \mid \theta^{(m)}) = \underbrace{\sum_k \gamma_1^{(m)}(k) \log \pi_k}_{\text{initial state}}$$

$$+ \underbrace{\sum_{t=1}^{T-1} \sum_{i,j} \xi_t^{(m)}(i,j) \log a_{ij}}_{\text{transitions}}$$

$$+ \underbrace{\sum_{t=1}^{T} \sum_k \gamma_t^{(m)}(k) \log b_k(x_t; \eta_k)}_{\text{emissions}}.$$

E-step by recursion

The full posterior assigns probabilities to K^T paths.

But Q requires only

$$\gamma_t^{(m)}(k) = \mathbb{P}_{\theta^{(m)}}(Z_t = k \mid x_{1:T})$$

and

$$\xi_t^{(m)}(i,j) = \mathbb{P}_{\theta^{(m)}}(Z_t = i, Z_{t+1} = j \mid x_{1:T}).$$

Forward–backward computes these marginals exactly.

For the derivation, fix θ and suppress the iteration index.

The forward quantity

$$\alpha_t(k) := p_\theta(x_{1:t}, Z_t = k).$$

It summarizes the observed past and the current state.

What is $\alpha_1(k)$?

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What is $\alpha_1(k)$?

$$\boxed{\alpha_1(k) = \pi_k b_k(x_1).}$$

$\alpha_t(k)$ is a joint likelihood, not a normalized posterior.

Derive the forward step

$$
\begin{aligned}
\alpha_t(j) &= p_\theta(x_{1:t}, Z_t = j) \\
&= \sum_i p_\theta(x_{1:t}, Z_t = j, Z_{t-1} = i) \\
&= \sum_i p_\theta(x_t, Z_t = j, x_{1:t-1}, Z_{t-1} = i) \\
&= \sum_i p_\theta(x_t \mid Z_t = j) P_\theta(Z_t = j \mid Z_{t-1} = i) p_\theta(x_{1:t-1}, Z_{t-1} = i)
\end{aligned}
$$

$$
\boxed{\alpha_t(j) = b_j(x_t) \sum_i \alpha_{t-1}(i) a_{ij}.}
$$

Recover the likelihood

At the final time, sum out the final state:

$$\boxed{p_\theta(x_{1:T}) = \sum_k \alpha_T(k).}$$

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At each time: K destination states, each with a sum over K origins.

Dense transition matrix: $O(TK^2)$.

Combine prefixes with the same endpoint instead of listing every path.

Filtering uses the past

Normalize the forward quantities:

$$\mathbb{P}_\theta(Z_t = k \mid x_{1:t}) = \frac{\alpha_t(k)}{\sum_j \alpha_t(j)}.$$

Is this the $\gamma_t(k)$ needed for the E-step?

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Is this the $\gamma_t(k)$ needed for the E-step?

The E-step conditions on $x_{1:T}$, including future observations.

The backward quantity

Define

$$\beta_t(i) := p_\theta(x_{t+1:T} \mid Z_t = i).$$

It summarizes the future observations given the current state.

We can see that

$$p_\theta(x_{t+1:T} \mid Z_t = i) = p_\theta(x_{t+1:T} \mid Z_t = i, x_{1:t})$$

Thus

$$p_\theta(x_{1:T}, Z_t = i) = \underbrace{p_\theta(x_{1:t}, Z_t = i)}_{\alpha_t(i)} \underbrace{p_\theta(x_{t+1:T} \mid Z_t = i)}_{\beta_t(i)}$$

Derive the backward step

Let

$$\beta_T(i) = 1 \qquad \text{for all } i = 1, \ldots, K.$$

Which next state could generate the remaining observations?

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&= \sum_j p_\theta(x_{t+1}, x_{t+2:T}, Z_{t+1} = j \mid Z_t = i) \\
&= \sum_j p_\theta(x_{t+2:T} \mid Z_{t+1} = j) p_\theta(x_{t+1} \mid Z_{t+1} = j) p_\theta(Z_{t+1} = j \mid Z_t = i)
\end{aligned}$$

$$\boxed{\beta_t(i) = \sum_j \beta_{t+1}(j) b_j(x_{t+1}) a_{ij}.}$$

Past meets future

Conditional on $Z_t = k$, past and future responses separate:

$$p_\theta(x_{1:T}, Z_t = k) = \alpha_t(k)\beta_t(k).$$

Now apply Bayes' rule.

$$\boxed{\gamma_t(k) = \frac{\alpha_t(k)\beta_t(k)}{p_\theta(x_{1:T})}.}$$

$$p_\theta(x_{1:T}) = \sum_j \alpha_t(j)\beta_t(j) \quad \text{for every } t.$$

A posterior transition

$$
\begin{aligned}
&p_\theta(x_{1:T}, Z_t = i, Z_{t+1} = j) \\
&\quad = p_\theta(x_{1:t}, x_{t+1}, x_{t+2:T}, Z_t = i, Z_{t+1} = j) \\
&\quad = \underbrace{\alpha_t(i)}_{\text{past}} \quad \underbrace{a_{ij}}_{\text{transition}} \quad \underbrace{b_j(x_{t+1})}_{\text{next response}} \quad \underbrace{\beta_{t+1}(j)}_{\text{future}}.
\end{aligned}
$$

Therefore

$$
\boxed{\xi_t(i,j) = \frac{\alpha_t(i) a_{ij} b_j(x_{t+1}) \beta_{t+1}(j)}{p_\theta(x_{1:T})}.}
$$

Check the marginals

$$\sum_k \gamma_t(k) = 1.$$

What happens when we sum $\xi_t(i,j)$ over one state?

$$\sum_j \xi_t(i,j) = \gamma_t(i), \qquad \sum_i \xi_t(i,j) = \gamma_{t+1}(j).$$

Summing a pair probability recovers its one-state marginal.

Update the initial law

Maximize over the probability simplex:

$$\max_{\pi} \sum_{k} \gamma_1^{(m)}(k) \log \pi_k, \qquad \pi_k \geq 0, \quad \sum_{k} \pi_k = 1.$$

How does this compare with the mixture weight update?

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$$\max_{\pi} \sum_{k} \gamma_1^{(m)}(k) \log \pi_k, \qquad \pi_k \geq 0, \quad \sum_{k} \pi_k = 1.$$

How does this compare with the mixture weight update?

$$\boxed{\pi_k^{(m+1)} = \gamma_1^{(m)}(k).}$$

Only the first latent state contributes to the initial distribution.

Optimize a transition row

Define

$$C_{ij}^{(m)} = \sum_{t=1}^{T-1} \xi_t^{(m)}(i,j)$$

For a fixed origin state i,

$$\max_{a_{i1},\ldots,a_{iK}} \sum_j C_{ij}^{(m)} \log a_{ij}, \qquad \sum_j a_{ij} = 1.$$

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The row-wise Lagrangian is

$$\mathcal{L}_i = \sum_j C_{ij}^{(m)} \log a_{ij} + \nu_i \left(1 - \sum_j a_{ij}\right).$$

The transition update

Differentiate with respect to a_{ij}.

$$\frac{\partial \mathcal{L}_i}{\partial a_{ij}} = \frac{C_{ij}^{(m)}}{a_{ij}} - \nu_i = 0.$$

$$1 = \sum_j a_{ij} = \frac{\sum_j C_{ij}^{(m)}}{\nu_i} \quad \Longrightarrow \quad \nu_i = \sum_j C_{ij}^{(m)}.$$

$$\boxed{a_{ij}^{(m+1)} = \frac{\sum_{t=1}^{T-1} \xi_t^{(m)}(i,j)}{\sum_{t=1}^{T-1} \gamma_t^{(m)}(i)}.}$$

Expected transitions divided by expected departures.

Fit weighted emissions

For each state,

$$\eta_k^{(m+1)} \in \arg\max_{\eta_k} \sum_{t=1}^{T} \gamma_t^{(m)}(k) \log b_k(x_t; \eta_k).$$

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- The weights are fixed throughout the M-step.
- Gaussian and Poisson models give closed-form updates.
- Other emission families may need numerical optimization.

The Baum–Welch loop

1. Initialize valid π, A, and distinct emission parameters.
2. **E-step:** compute γ, ξ by forward–backward.
3. **M-step:** update π, A, and the weighted emissions.
4. Evaluate the new log-likelihood by a forward pass; repeat.

Stop when the improvement is small; check for likelihood decreases.

Avoid numerical underflow

Recall that:

$$\alpha_t(j) = b_j(x_t) \sum_i \alpha_{t-1}(i) a_{ij}.$$

When $t \to \infty$, $\alpha_t(j) \to 0$. Long products can become too small to represent.

Solution: Use the log-sum-exp identity:

$$\log \sum_i e^{u_i} = c + \log \sum_i e^{u_i - c}, \qquad c = \max_i u_i.$$

Subtract the maximum before exponentiating.

Gaussian responses

$$X_t \mid Z_t = k \sim \mathcal{N}(\mu_k, \sigma_k^2).$$

The E-step uses the current emission log densities:

$$\log b_k(x_t) = -\frac{1}{2}\log(2\pi\sigma_k^2) - \frac{(x_t - \mu_k)^2}{2\sigma_k^2}.$$

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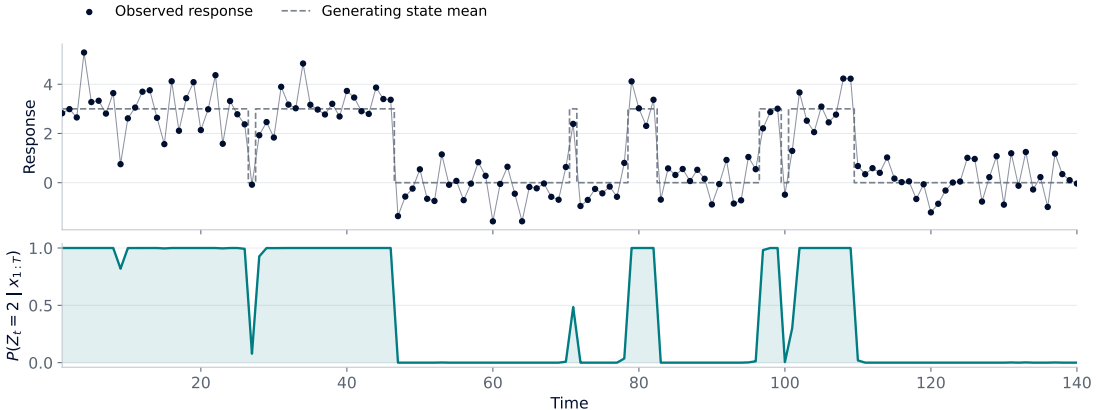
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Illustrative states:

$$(\mu_1, \mu_2) = (0, 3), \qquad (\sigma_1^2, \sigma_2^2) = (0.49, 0.64).$$

Gaussian state inference



Smoothing uses the whole sequence to infer the state at each time.

Simulated data. Posterior probabilities are computed under the fitted HMM.

The Gaussian objective

For state k, write $w_t = \gamma_t^{(m)}(k)$ and $N_k = \sum_t w_t$.

$$Q_k(\mu_k, v_k) = -\frac{N_k}{2} \log v_k - \frac{1}{2v_k} \sum_t w_t (x_t - \mu_k)^2 + C,$$

where $v_k = \sigma_k^2$ and C is constant in μ_k, v_k.

This is the same weighted Gaussian fit as in mixture EM.

Update the state means

Differentiate the weighted objective with respect to μ_k.

$$\frac{\partial Q_k}{\partial \mu_k} = \frac{1}{v_k} \sum_t w_t (x_t - \mu_k) = 0.$$

$$\boxed{\mu_k^{(m+1)} = \frac{\sum_t \gamma_t^{(m)}(k) x_t}{\sum_t \gamma_t^{(m)}(k)}.}$$

Update the variances

Use the new mean with the old posterior weights:

$$S_k = \sum_t \gamma_t^{(m)}(k)(x_t - \mu_k^{(m+1)})^2.$$

$$\frac{\partial Q_k}{\partial v_k} = \frac{S_k - N_k^{(m)} v_k}{2v_k^2} = 0.$$

$$\boxed{(\sigma_k^2)^{(m+1)} = \frac{S_k}{N_k^{(m)}}.}$$

Which state path is best?

Marginal decoding chooses each state separately:

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Viterbi decoding chooses the most probable *whole path*:

$$\widehat{z}_{1:T}^{\mathrm{Vit}} \in \arg\max_{z_{1:T}} p_\theta(z_{1:T} \mid x_{1:T}) = \arg\max_{z_{1:T}} p_\theta(x_{1:T}, z_{1:T}).$$

Fix the fitted parameters before decoding.

Keep the best prefix

Define the best partial path ending in state j:

$$\delta_t(j) := \max_{z_{1:t-1}} p_\theta(x_{1:t}, Z_{1:t-1} = z_{1:t-1}, Z_t = j).$$

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Forward: $\alpha_t(j)$ Sum the scores of all prefixes.

Viterbi: $\delta_t(j)$ Keep only the best prefix score.

The Viterbi recursion

Initialize:

$$\delta_1(j) = \pi_j b_j(x_1).$$

How do we extend the best prefix to time t?

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=&\max_i \max_{z_{1:t-2}} p_\theta(x_{1:t-1}, x_t, Z_{1:t-2} = z_{1:t-2}, Z_{t-1} = i, Z_t = j) \\
=&p_\theta(x_t \mid Z_t = j) \max_i \left\{ p_\theta(Z_t = j \mid Z_{t-1} = i) \max_{z_{1:t-2}} p_\theta(x_{1:t-1}, Z_{1:t-2} = z_{1:t-2}, Z_{t-1} = i) \right\}
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\end{aligned}$$

$$\boxed{\delta_t(j) = b_j(x_t) \max_i \{a_{ij}\delta_{t-1}(i)\}.}$$

Replace the forward sum by a maximum.

Remember the predecessor

The best score alone does not tell us the entire path.

$$\psi_t(j) \in \arg\max_i \{\delta_{t-1}(i) a_{ij}\}.$$

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$$\psi_t(j) \in \arg\max_i \{\delta_{t-1}(i) a_{ij}\}.$$

$\psi_t(j)$ records which previous state led to the best score at j.

Store these backpointers for $t = 2, \ldots, T$.

Recover the whole path

Choose the best final state:

$$\widehat{z}_T \in \arg\max_j \delta_T(j).$$

Recover the whole path

Choose the best final state:

$$\widehat{z}_T \in \arg\max_j \delta_T(j).$$

Follow the stored predecessors backward:

$$\boxed{\widehat{z}_t = \psi_{t+1}(\widehat{z}_{t+1}), \qquad t = T-1, \ldots, 1.}$$

One final choice, then backtracking through the sequence.

Three distinct tasks

Forward–backward	Posterior state and transition probabilities.
EM / Baum–Welch	Estimate parameters using those posterior expectations.
Viterbi	Find one most probable complete state path.

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Forward–backward	Posterior state and transition probabilities.
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Viterbi	Find one most probable complete state path.

Average over paths to fit with EM; maximize over paths to decode.